Homework 8

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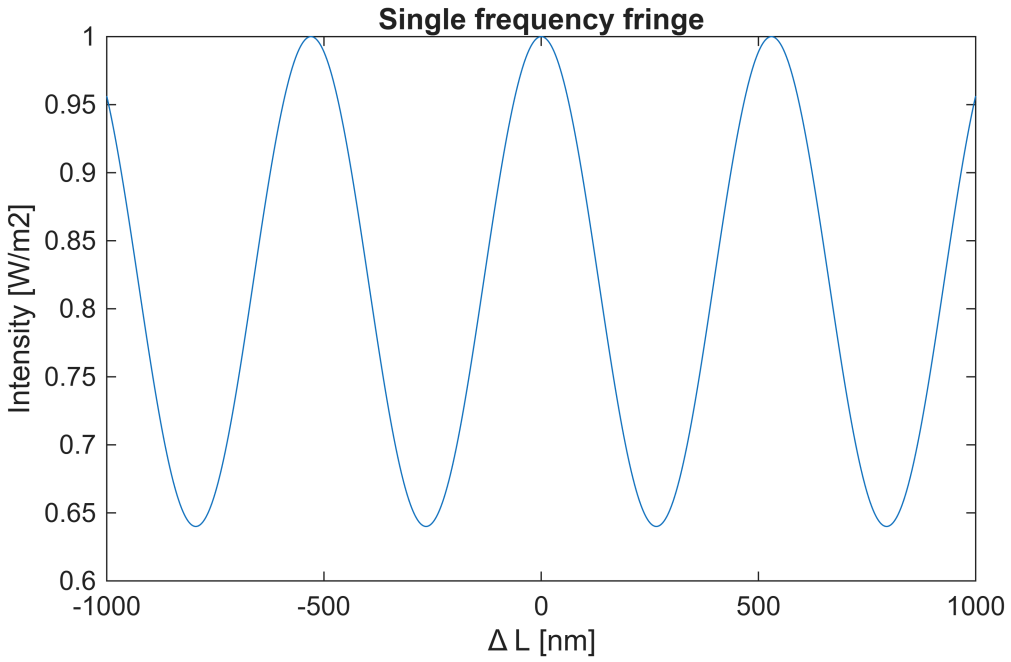
Part 1

Problem 9.3

```
% Givens
E_R0 = 0.1; % [W/m2]
E_S0 = 0.9; % [W/m2]
lambda = 1060; % [nm]
k_S = 2*pi/lambda; % [rad/nm]
k_R = 2*pi/lambda; % [rad/nm]
l_R = 1000000; % [um] Reference arm length (1m)
Delta_l = linspace(-1000,1000, 1000); % [nm]
l_S = l_R + Delta_l; % [um] Sample arm length

I = E_R0^2 + E_S0^2 + 2*E_S0*E_R0*cos(2*k_S*l_S - 2*k_R*l_R);

plot(Delta_l, I)
xlabel("\Delta L [nm]")
ylabel("Intensity [W/m2]")
title("Single frequency fringe")
```




```
I_avg = E_R0^2 + E_S0^2;
```

```
max_contrast = 2*E_S0*E_R0 / I_avg
```

```
max_contrast =
0.2195
```

Problem 9.7

```
% Constants
modulation = 0.05;
I_DC = 1;
I_AC = modulation * I_DC;
num_measurements = 1000;
num_freq = 1000;

lambda_0 = 830; % [nm] Center wavelength

bandwidth = [100,50,25]; % [nm]
backscatterer_separation = [4, 2, 1, 0.5]; % [Multiples of l_c]

tl = tiledlayout(numel(bandwidth), numel(backscatterer_separation));

for Delta_lambda = bandwidth

    l_c = 4*log(2)/pi*lambda_0^2/Delta_lambda % [nm] Coherence length

    Delta_l = linspace(-l_c*4, l_c*4, num_measurements);
    l_freq = num_measurements/8*l_c; % [1/nm] Spatial frequency of Delta_l
measurements

    for n = backscatterer_separation

        separation = n*l_c;

        location_front = -n*l_c/2; % [nm]
        location_back = n*l_c/2; % [nm]

        I_front = 0 * Delta_l;
        I_back = 0 * Delta_l;

        for lambda = linspace(lambda_0-Delta_lambda/2, lambda_0+Delta_lambda/2, 100)

            k = 2*pi/lambda; % [1/nm]

            I_front = I_front + I_AC/2/num_measurements*cos(2*k*(Delta_l -
location_front));
            I_back = I_back + I_AC/2/num_measurements*cos(2*k*(Delta_l -
location_back));

        end
```

```
        I = I_front + I_back + I_DC; % Raw interferogram

        I_rect = abs(I - mean(I)); % Rectify

        I_fft = fft(I_rect, num_freq); % Lowpass filter
        I_fft(15:num_freq-15) = 0;
        I_low = real(ifft(I_fft, num_measurements));

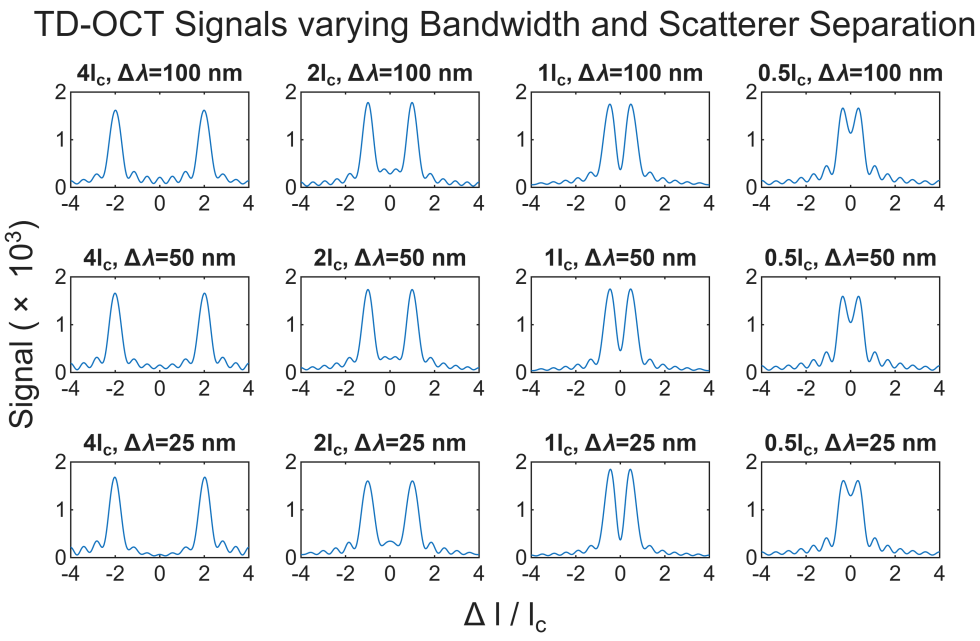
        nexttile
        plot(Delta_l/l_c, I_low*1e3)
        ylim([0,2])
        xlim([-4,4])
        title(sprintf("%gl_c, \\Delta\\lambda=%g nm", n, Delta_lambda))

    end

end
```

```
l_c =
6.0798e+03
l_c =
1.2160e+04
l_c =
2.4319e+04
```

```
title(tl, "TD-OCT Signals varying Bandwidth and Scatterer Separation")
xlabel(tl, "\Delta l / l_c")
ylabel(tl, "Signal (\times 10^3)")
```



Part 4

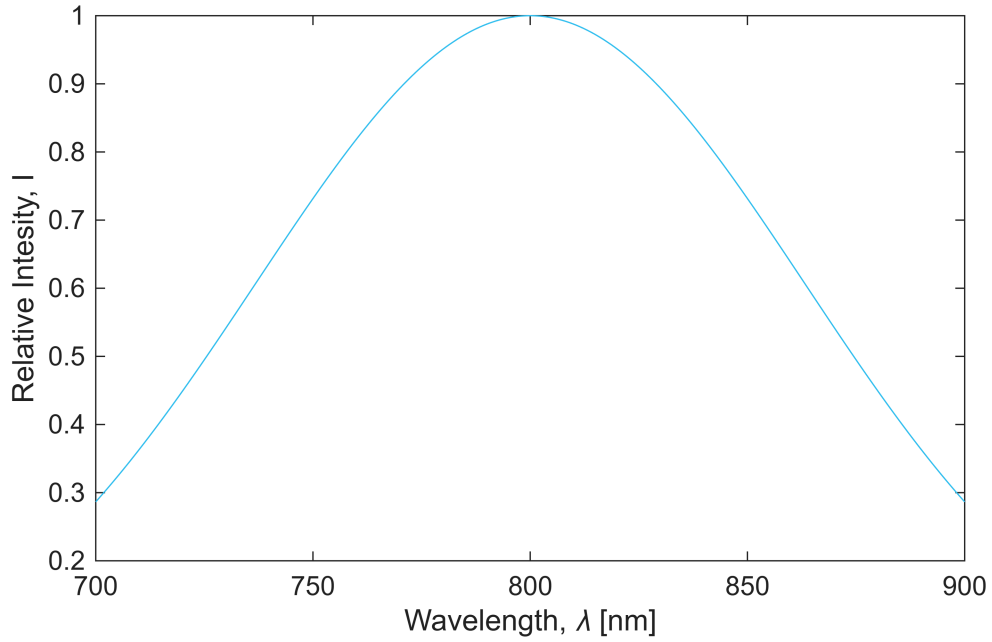
```
Delta_l = linspace(-100e3,100e3,1000); % [nm]
lambda = linspace(700,900,1000); % [nm]

[L, Y] = meshgrid(Delta_l, lambda);

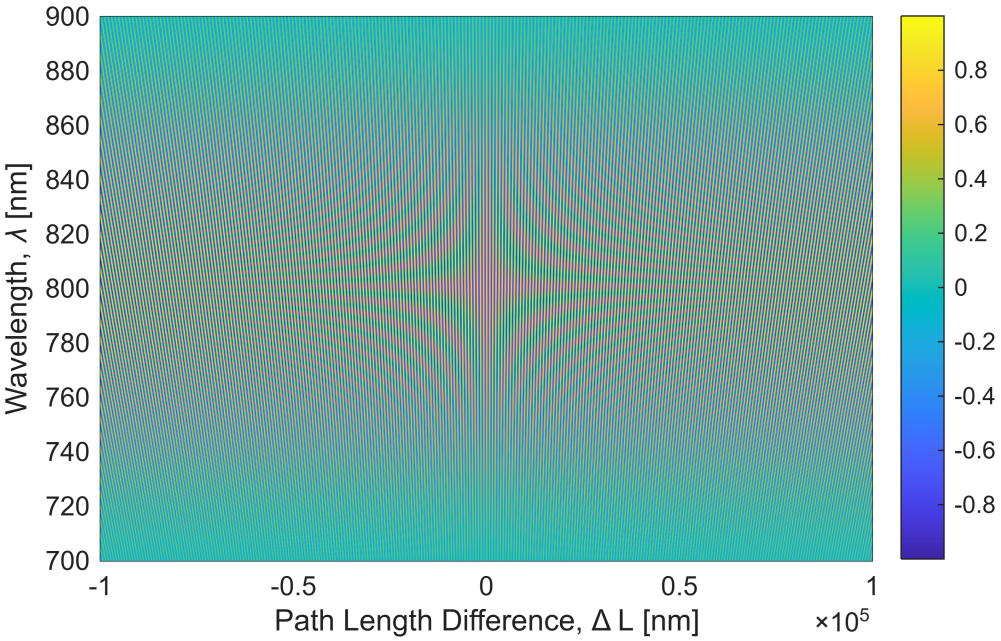
I_lambda = exp(-(Y-800).^2/8000);

I = I_lambda.*cos(2*pi*L./Y);

figure()
plot(lambda, I_lambda)
xlabel("Wavelength, \lambda [nm]")
ylabel("Relative Intesity, I")
```




```
figure()
h = pcolor(L, Y, I);
set(h, 'EdgeColor', 'none');
view(2);
xlabel("Path Length Difference, \Delta L [nm]")
ylabel("Wavelength, \lambda [nm]")
colorbar()
```


900
880
860
840
820
800
780
760
740
720
700
Wavelength, λ [nm]
-1
-0.5
0
0.5
1
Path Length Difference, Δ L [nm]
×10^5
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8